

EXAM 7 – SPRING 2012

23. (2 points)

a. (1.5 points)

Describe three main sources of underwriting risk that should be modeled separately in an insurance company's internal solvency model.

b. (0.5 point)

Explain the difficulty in determining the cross-class correlation component of the internal model.

CONTINUED ON NEXT PAGE

Question 23 Sample Answer

Solution 1

a)

1. Attrition losses – this is the risk that there will be more claims than expected, or claims will be more severe than expected.
2. Large losses – the risk that there will be more large losses or larger losses than expected.
3. Catastrophe risk – the risk that a single event will cause many independent risks to become correlated.

Solution 2

a)

1. Attritional claims need to be modeled – this is the risk of having many (high frequency) small (low severity) claims.
2. Large claims need to be modeled. These are the larger claims that have the potential to cause large unexpected losses that the company needs to be able to survive.
3. Catastrophe claims must be modeled. These are low frequency, high severity stress losses that occur in the extreme tail of the loss distribution. These can cause insolvency quickly if a company is not prepared.

Solution 3

a)

1. Pricing – will premiums be enough to cover business written/ profit
2. Claims – will you have more claims than expected or perhaps a higher number of large claims or just higher than expected severity on small claims
3. Product design risk – will coverage be interpreted to be more broad than anticipated? Exposing one to unanticipated loss types or latent exposures?

Solution 1

- b) This is difficult to determine due to the nature of cross class correlation during extreme scenarios. There is very little, if any, data available illustrating extreme events and scenarios. This causes the correlation to primarily be based on actuarial judgment.

Solution 2

b) Correlations may be stronger in the tails of distributions. The problem is: The lack of data in tails within the company to estimate appropriately & verify. Need to use:

- Industry/expert advice
- Copulas or other technical theory to model structural dependencies not observed yet.

Examiner Comment

For the a. part, alternate solutions:

- Latent risk
- Correlation
- Expenses

Common mistakes:

- Not explaining the risk
- Several candidates did not explain the risk but how they would change the model. That is not what the question asked.

For the b. part, credit was lost for not explaining enough. For example, saying it would be difficult in the tail, but not explaining why (data) would result in partial credit.